

Psychological barriers to participation in the labor market: policy brief

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1. Introduction and research question

This study examines the context of rural Ghana, specifically the region north of Tamale. The local population is heavily reliant on agriculture, and the poverty rate in this area is remarkably high, with over 70 percent of people classified as multi-dimensionally poor across the study districts (Ghana Statistical Service, 2023).

The study, conducted between March and September 2024, included six rounds of post-baseline data collection. This timeline is crucial as it corresponds to the "lean season"—the period after planting and before the harvest of the major staple grain, maize. This season is characterized by limited work opportunities (especially wage labor), severe food insecurity, low income, and high psychological stress.

This research is part of a special issue published in the *Journal of Development Economics* focusing on "Mental Health and Socioeconomic Outcomes." It investigates the multifaceted connections between mental health and various aspects of socioeconomic well-being, specifically exploring how mental health intersects with economic factors, labor market participation, productivity, and social integration.

Anxiety and depression in the rural Ghana setup can not only be an effect, but also a cause, creating a loophole. Poor mental health symptoms like avoidance behaviors, low energy levels, anhedonia, nervousness, are likely to affect decisions of agents in the labor market (Ridley et al. 2020).

One of the authors, Crick Lund, is also one of the lead authors of "Poverty and common mental disorders in low and middle income countries: A systematic review (2010)", a research conducted on 115 community-based studies across low and middle-income countries. It reveals that food insecurity in poorer countries drives mental illness much more strongly than a simple lack of monetary income.

Most of the economic debate has always been committed to the influence of mental illness on productivity on the job. Carvalho et al. instead are focusing on how psychological factors can impact labor force's agents in decisions related to job take-up.

This is consistent with a new streak in economics. An example of that is the Madeline McKelway (2021) study on women's participation in the job market in rural Uttar Pradesh (India). It reports the experiment made by offering women a psychosocial intervention to raise generalized self-efficacy (GSE), finding out that the treatment increased the short-run female employment. The key finding concludes that the agency of women in poor countries may be constrained by women's own sense of agency, because of their life experiences: low education, regressive gender attitudes, and other inequities they face (Duflo, 2012; Jayachandran, 2015).

The core contribution of Carvalho et al. is providing a deeper understanding of how mental health conditions act as barriers *before* a worker even enters the labor market. In the observed region, the prevalence of these conditions is severe: Ae-Ngibise et al. (2023) observe a probable depression rate of 15.6%, and Barker et al. (2022) note that about 58% of the general population suffers from symptoms of psychological distress—a staggering figure compared to approximately 13% in the US.

The research questions of this study are broken down into two phases:

1. Knowing that anxiety and depression hinder women from poor households from participating in the labor market, can working from home soften this effect in terms of willingness to work and effectively influence job take-up decisions?
2. Do women who suffer from anxiety and depression, but are willing to work from home, exhibit differences in terms of labor supply, output, earnings, and quit rates compared to those without mental health issues?

If policy makers assume that the only obstacles faced by people in poverty state are technical or financial, without considering the weight of mental health, they fail to propose feasible and participated solutions, because they ask vulnerable people to overcome a logistical and social hurdles (leaving the house, interacting with others) that their condition makes insurmountable.

2. Empirical strategy and data

The identification strategy used is the randomized controlled trial. Once established that mental health conditions significantly reduce a person's willingness to accept traditional jobs outside the home, the conclusion is that if the authors only observed people naturally working in the labor market as it is, the data would be heavily biased because people with poor mental health naturally select out of the workforce.

By randomly offering work-from-home positions to women who had already stated they were willing to work from home, selection bias effects are avoided.

Offering identical jobs lets also to bypass the phase of job searching for participants. This helps because differences in how people look for work can be heavily influenced by depression or anxiety.

The fact that the jobs are identical offers also the possibility to hold the characteristics of the job constant, to prevent unobservable features of different jobs from unobservably and endogenously changes, according to the individual's characteristics such as their mental health status.

The objective is to create a neutral environment (or as neutral as possible) where they could isolate the effect of the job offer on their outcomes.

The randomization allows the researchers to decompose the timeline of the employment and the process is split into two steps: the initial decision of accepting the job and the actual work, producing output, earning money and not quitting.

The research process and the empirical strategy are split into two steps:

Step 1: Baseline Survey and Willingness to Work

The first step was selecting households using a random walk procedure in 30 communities (villages). The eligibility criteria for a household were access to a phone and having five or fewer adults with a total household size of less than 16 members. This resulted in 838 eligible households, of which 491 had direct female respondents who were asked to provide basic socio-demographic information.

The researchers described a job offer (stitching bags) and asked if the women would be willing to do such work, both from home and outside of the home. To address the mental health aspect, participants responded to the PHQ-2 screening tool (assessing depression) and the GAD-2 module (assessing anxiety). Mental health is then measured as the standardized average of these two variables.

To analyze the observational relationship between mental health conditions and willingness to work, the following linear probability model is estimated using ordinary least squares (OLS):

$$Y_i = \alpha_0 + \alpha_1 \text{Mental health}_i + X_i \alpha_2 + \epsilon_i$$

Where:

Y_i : binary, dummy, variable equal to one if the respondent is willing to work and zero otherwise;

Mental health_i : PHQ-2 or GAD-2 standardized or a standardized average of both;

X_i : a vector of controls, including the order of the willingness to work questions (which was randomized), whether the household head or his spouse were both present or not (which influenced the selection of the respondent, because if both the household head and his spouse are present, one of them is randomly selected to be the main respondent). If either the head or his spouse are absent, the enumerator randomly selects one of the present adult household members to respond. Other controls are the household size and number of adults, age of the respondent, and marital status.

Table A: Association between alternate mental health measures and work offer acceptance with controls.

	(1)	(2)	(3)
Willingness to work outside			
High PHQ-2	-0.039 (0.029)		
High GAD-2		-0.049* (0.029)	
High PHQ-2 and GAD-2			-0.078** (0.035)
Dep. Var. Mean	0.89	0.89	0.89
Willingness to work from home			
High PHQ-2	0.002 (0.022)		
High GAD-2		0.027 (0.021)	
High PHQ-2 and GAD-2			0.003 (0.024)
Dep. Var. Mean	0.95	0.95	0.95
p-value: W.f. Home == W. outside	0.10	0.0030	0.0072
Observations	491	491	491

Carvalho et al. (2026)

These women were also asked if there was another woman or another adult in general in the household with willingness to work, arriving to a total of 1567 women. Of these, 1447 are willing to work from home.

The authors adopted this sampling procedure because the current research served as a pilot study for a broader randomized controlled trial that might include men the following year. Widening the study could reveal if and how anxiety and depression might affect their job take-up decisions differently than women. In this region, the majority of income typically comes from male adults, who earn more per hour and work outside of the home more frequently. The study could highlight gender differences and if the work-from-home solution is universally effective.

To provide a boost on male labor market participation could also be a measure of poverty alleviation for the families considered. It's plausible that, if men are the primary earners and their mental health conditions are keeping them out of the labor market, then proposing alternative solutions could increase the total household income significantly.

Severe mental health diseases of a person can impact all their household. We can not exclude that there could be spillover effects. The evidence from the present study suggests that a woman's poor mental health makes her less likely also to sign up other household members for work. If men are included in the analysis, researchers can observe the full household dynamic.

To ensure the validity of the statistical inference against heteroskedasticity, the first model estimates use robust standard errors. Additionally, in Table 2, when analyzing the specific reasons for work refusal across pooled locations, standard errors are clustered at the individual level to account for the fact that a single respondent could provide multiple reasons for rejecting work.

The coefficient α_1 is reported in its values for different measures of mental health in Table 1.

Table 1: Correlation between mental health and willingness to work.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Willingness to work outside the home							
PHQ-2 (std)	-0.030* (0.017)	-0.030* (0.018)	-0.035 (0.024)	-0.050* (0.028)	-0.038 (0.024)	-0.039 (0.024)	-0.055* (0.029)
GAD-2 (std)	-0.029* (0.015)	-0.025 (0.016)	-0.054** (0.022)	-0.063** (0.025)	-0.055** (0.024)	-0.058** (0.025)	-0.071*** (0.027)
Ave. of PHQ-2 (std) and GAD-2 (std)	-0.037** (0.019)	-0.035* (0.020)	-0.053** (0.025)	-0.069** (0.030)	-0.056** (0.026)	-0.059** (0.027)	-0.077** (0.031)
Dep. Var. Mean	0.89	0.89	0.91	0.91	0.91	0.91	0.91
Willingness to work from home							
PHQ-2 (std)	0.004 (0.012)	0.003 (0.012)	0.008 (0.008)	0.010 (0.010)	0.007 (0.007)	0.008 (0.008)	0.008 (0.008)
GAD-2 (std)	0.011 (0.012)	0.012 (0.012)	0.005 (0.005)	0.005 (0.005)	0.005 (0.005)	0.006 (0.006)	0.003 (0.004)
Ave. of PHQ-2 (std) and GAD-2 (std)	0.010 (0.014)	0.009 (0.014)	0.008 (0.008)	0.010 (0.010)	0.007 (0.007)	0.009 (0.009)	0.007 (0.008)
Dep. Var. Mean	0.95	0.95	0.99	0.99	0.99	0.99	0.99
Observations	491	482	156	156	156	156	156

Carvalho et al. (2026)

Panel A presents results about working from a local worksite decisions and Panel B presents results about working from home. All regressions use robust standard errors.

The equation is estimated separately based on willingness to work from home or to work outside. Ten households of these 400 were unavailable because those families left the village

in the meantime. They did not even receive the job offer, so they have been canceled out of the sample without bias problems.

The researchers randomly selected 400 households among the households with at least one woman willing to work from home. Then, if the household had more than a woman, they randomly selected only one woman.

Finally, the sample is reduced to 390 households, with 390 corresponding women.

The treatment group, composed by women who received the job offer, is of 320 units. The control group is 70 units.

Step 2: The Randomized Controlled Trial (RCT)

After establishing the baseline, the researchers implemented the RCT. The treatment consisted of actually offering the bag-stitching job to be performed from home.

From the broader sample of women willing to work from home, the researchers randomly selected 400 households. Ten households were unavailable as they had left the village, reducing the sample to 390 women. Among these, the treatment group (who received the actual job offer) consisted of 320 units, while the control group (who did not receive the offer) consisted of 70 units.

The objective of this phase is to determine if depressed/anxious women working from home have differences in labor supply, output, earning, and quit rates compared to non-depressed/non-anxious women. To capture the causal effects of the treatment and its interaction with mental health, the following regression is estimated:

$$Y_{i,t} = \gamma_0 + \gamma_1 \text{job offer}_i + \gamma_2 \text{Mental health}_{i,0} + \gamma_3 \text{Mental health}_{i,0} * \text{job offer}_i + T_t + X_i \gamma_4 + \zeta_{i,t}$$

where:

$Y_{i,t}$ is the outcome Y for individual i in period t;

job offer is equal to one if the individual i was randomly offered a job;

mental health is one of the variables measured in the baseline (period 0);

T_t are period fixed effects;

X_i is the vector of controls;

$\zeta_{i,t}$ is the error for individual i at time t.

$\gamma_3 \text{Mental health}_{i,0} * \text{job offer}_i$ is the interaction term. It is capturing that the impact of the job offer on the dependent variable is conditional to the individual baseline mental health score. If the regression shows that the coefficient γ_3 is statistically significant, it proves that the impact of the treatment varies depending on mental health conditions. Standard errors are clustered at the household level.

3. Empirical results and findings

As seen in table 1, mental health acts as a severe barrier to traditional job take-up. The linear probability models reveal a strong negative correlation between poor mental health scores and the willingness to work outside the home. For instance, individuals suffering from both anxiety and depression, (measured as the standardized average of the two), are significantly more likely to refuse a job at a common worksite.

In the top panel, an increase of 1 in standard deviation of the combined average index yields a 3.7 percentage point reduction of the willingness of accepting work outside the home in the baseline model. This result stays stable and statistically significant even in the robust model that includes every control, where there is a 7.7 percentage point reduction.

However, in the bottom panel this negative correlation disappears and the analysis of the coefficient of interest does not show a statistically significant correlation between mental health problems and the decision of accepting the job from home. The most important point is the interaction term, which is small and never statistically significant.

Most importantly, when the offered job is home-based, mental health conditions do not stop participation, proving that the psychological barrier is actively interfering with women's choices regarding mobility and social exposure in the labor market.

The initial barrier is the primary driver of labor market disparities for this demographic group.

Women that were not willing to work at all were asked to provide the reason for their refusal, and the results of this correlation are expressed in table 2.

People with higher depression and anxiety scores are significantly less likely to say that they already have paid work as a reason for refusing job offers.

Instead, an increase of 1 in standard deviation of PHQ-2 (depression) increases by 8.9 percentage points the possibility of citing inability or unwillingness to work as a reason to refuse the job.

Table 2: Association between depression and reasons for work refusal

	(1) Have work	(2) Household work	(3) Unable or unwilling	(4) Student	(5) Transportation
PHQ-2 (std)	-0.087** (0.033)	0.008 (0.044)	0.089* (0.048)	-0.023 (0.020)	0.023 (0.040)
GAD-2 (std)	-0.094** (0.039)	0.090 (0.055)	0.051 (0.061)	-0.013 (0.016)	-0.008 (0.036)
Ave. of PHQ-2 (std) and GAD-2 (std)	-0.102** (0.039)	0.050 (0.053)	0.082 (0.058)	-0.021 (0.020)	0.012 (0.043)
Reason share	0.114	0.380	0.367	0.0253	0.115
Observations	79	79	79	79	52

Carvalho et al. (2026)

According to the authors, depression and anxiety are significant predictors of labor supply, through reduced willingness to accept work opportunities when the work requires to leave home, though not when the work is offered at home.

Practically, table 3 presents the results of the second phase, which is the randomized controlled trial where a randomly selected group of willing participants was offered the job of

stitching bags from home. Receiving the random job offer increases participation, the number of days worked, and overall income.

Panel 1, 2 and 4 evaluate the entire sample, with a total of 2188 panel data, to compare treatment and control group. Panel 3 measures productivity, the outcome is the number of bags stitched, so it evaluates only those who worked, with a total of 1920 observations.

Analyzing the statistically significant coefficients, first of all we can see the absolute impact of the job offer. Being randomly offered a job increased a woman's probability of working by 43.5 percentage points against a baseline control group mean of just 17.2.

The job offer increased the number of days worked by 1.69 days out of a 10-day window (control mean of 1.05 days). The bi-weekly income is boosted by 103.24 GHS (14.97 as average in the control group).

The most crucial econometric finding of the paper lies in the interaction terms between the baseline mental health measures and the job offer dummy (γ_3).

In labor economics, mental illness is traditionally modeled as a factor that depreciates human capital, leading to lower productivity and lower earnings. If this classical view held true, we would expect the interaction term to be negative and statistically significant, meaning that the positive effect of the job offer would be attenuated for women with severe depression or anxiety.

The interaction terms in panel 1, 2 and 4 are not statistically significant across all columns. For instance, looking at "Work Income", the interaction coefficients (-5.671 or -3.782) have large standard errors relative to their size, rendering them statistically indistinguishable from zero.

This proves that, once the work begins at home, women with anxiety and depression work, produce bags and earn amounts identically to their peers without mental health issues.

This kind of job offer is equally valuable to everyone.

In panel 3, isolating the treated individuals, it shows that baseline depression and anxiety have no statistically significant impact on the number of bags a woman completes.

From an analytical standpoint, this "null result" on the interaction term is exactly what proves the success of the intervention. It econometrically demonstrates that once the environmental stressor (working outside) is removed, women suffering from anxiety and depression work, produce, and earn at identical rates to their neurotypical peers. The work-from-home treatment successfully neutralizes the productivity penalty traditionally associated with poor mental health.

Table 3: Work offers, labor supply, income and mental health

	(1) –	(2) PHQ-8 (std)	(3) GAD-7 (std)	(4) Ave. of PHQ-8 (std) and GAD-7 (std)
Working				
Job offer	0.435*** (0.032)	0.432*** (0.032)	0.429*** (0.033)	0.429*** (0.032)
Baseline mental health measure		0.041 (0.026)	0.036 (0.026)	0.044 (0.027)
Baseline mental health measure × job offer		–0.036 (0.031)	–0.015 (0.030)	–0.028 (0.032)
Control Mean	0.172	0.172	0.172	0.172
Observations	2188	2188	2188	2188
Days worked				
Job offer	1.693*** (0.204)	1.678*** (0.205)	1.662*** (0.207)	1.666*** (0.206)
Baseline mental health measure		0.177 (0.191)	0.205 (0.176)	0.219 (0.198)
Baseline mental health measure × job offer		–0.141 (0.211)	–0.117 (0.193)	–0.144 (0.219)
Control Mean	1.052	1.052	1.052	1.052
Observations	2188	2188	2188	2188
Bags produced				
Baseline mental health measure		–0.032 (0.175)	0.171 (0.163)	0.084 (0.187)
Mean		8.998	8.998	8.998
Observations		1920	1920	1920
Work income				
Job offer	103.232*** (3.559)	103.040*** (3.602)	102.543*** (3.710)	102.714*** (3.664)
Baseline mental health measure		4.956 (3.934)	5.112 (4.063)	5.752 (4.357)
Baseline mental health measure × job offer		–5.671 (4.394)	–3.782 (4.449)	–5.378 (4.830)
Control Mean	14.97	14.97	14.97	14.97
Observations	2188	2188	2188	2188

Carvalho et al. (2026)

4. Conclusions and policy implications

Contrary to the existing literature that often links mental illness to lower on-the-job productivity in traditional settings (Ridley et al., 2020; Lund et al., 2024), this study finds no evidence that mental health diminishes output when the work environment accommodates the workers' psychological constraints.

The RCT proves that work-from-home arrangements in this setting successfully eliminate the productivity penalties traditionally associated with poor mental health.

Traditional cash-for-work programmes can result inefficient if the underlying implication is that economically disadvantaged people will automatically self-select into the labor market when they have the chance to do so, without considering starting conditions and psychological constraints.

The article is in the open knowledge repository of the World Bank, in the section dedicated to policy research about development issues. This inclusion suggests that its exposure could influence policy making and a change in the classic setting of cash-for-work programmes.

Although it is premature to quantify the paper's long-term academic contribution due to its recent publication, its inclusion in this special issue signals that the debate views its focus on the intersection of mental health and structural labor barriers as a vital new frontier for

development policy. As mentioned, a further contribution will be determined by the future expansion of the randomized controlled trial.

The line of thought that connects the treatment of poor mental health and policies for unemployment should also guide the path of public policymakers.

Work from home could be a useful tool to improve the productivity of those who choose not to participate in the traditional labor market because of mental health issues and could increase the labor supply.

Poverty is not only an economic phenomenon but also a social one that repeats itself if not addressed at its primary drivers. The comment by Nina Collavo from Cornell University about the present study on Ghanaian women pinpoints how unemployment itself is a mental health risk, potentially causing a person's unemployment-related mental illness to inhibit their ability to seek a new job (Yang, 2024).

If you can enter the labor market with a remote option, gain earnings, and your mental health improves, it'll put you in a much better place (Schofield, 2026).

While WFH opportunities provide a tool to bypass the immediate psychological barrier of job take-up, they cannot serve as a standalone cure for a nation's mental health crisis, especially in contexts of severe depression rate like rural Ghana, risking of becoming a way to structurally isolate people with mental health problems.

As the authors explicitly caution, traditional work outside of the home offers critical secondary benefits, such as social interaction and network building, which are vital therapeutic mechanisms for improving long-term mental health outcomes.

Furthermore, the policy-making process should not be confined to an economic paradigm that values mental health interventions primarily as an instrumental tool to extract labor from vulnerable populations and increase productivity. Instead, the primary goal should be restoring agency, dignity and quality of life of the people involved.

Work from home does not exempt countries' governments to provide a solid mental health support system. In the case of the study, Ghana Health Service collaborated and monitored.

True economic development requires state-level public policies to actively integrate complementary mental health interventions and screening services directly into their employment frameworks. Without a foundational healthcare system to treat the root causes of psychological distress, WFH initiatives remain a short-term economic bandage in a systemic problem that requires a permanent solution for human capital development.

Key strategies for governments to reach the integration are enabling support interventions, guaranteeing accessibility to the consultation of mental health professionals, ensuring that employment laws and actual conditions align with human rights instruments, and integrating mental health awareness into training programmes.

Integrated mental health and employment services can improve labour market outcomes for people with mental ill-health if implemented rigorously (Arends et al. 2014).

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